

Summary

Policy researcher and applied economist applying causal inference, experimentation, and large-scale data analysis to high-stakes decisions, from national policy at key government agencies in Chile to multi-year research programs at UC Berkeley. Designed and ran an \$880K FAIR/NHH-funded randomized controlled trial (RCT) to understand missing results in economics; led the Open Policy Analysis Initiative at BITSS, developing a framework for transparent policy analysis published in *Science & Public Policy*. Expertise using administrative and survey data for quasi-experimental impact evaluations (DD, RDD, IV) and policy analysis (microsimulations, cost-benefit analysis), with an emphasis on transparency and reproducibility. Proven track record building and leading teams, securing \$2M+ in competitive funding (FAIR/NHH, NIH, Templeton, Arnold Ventures, Coefficient Giving), and translating rigorous evidence into actionable recommendations for agencies including CBO, IADB, and USAID. Actively testing LLM-based tools to scale hypothesis encoding and exploring AI agents for reproducible research pipelines.

Job Market Paper

“Beyond Publication Bias: Characterizing and Understanding Missing Results in Economics” (with Miguel, E., Pathan, G., Rocha, V., Shaji, A., Sørensen, E., Tungodden, B.). [Latest version here](#).

Work Experience

2019–2026 **Assistant Project Scientist**, University of California, Berkeley

- **Co-PI, Publication Bias and Standardization of Results research program (2023–2026)**: secured \$880K (FAIR/NHH) to measure and explain under-reporting of results in economics. Designed and ran a randomized experiment with a sample of over 300 teams of academic researchers; built a hypothesis-encoding protocol and led a team of 3 research assistants. [Working paper](#).
- **Open Policy Analysis Initiative, BITSS (2019–2022)**: led program embedding open-science methods into government policy analysis. Presented the framework and delivered tailored recommendations to agencies including CBO, IADB, USAID, and Chile’s Ministry of Finance. Framework published in *Science & Public Policy*. bitss.org/opa/
- **Social Science Reproduction Platform**: co-designed and oversaw development of a platform for crowdsourcing computational reproducibility, which hosted 300+ reproductions by researchers worldwide (200+ papers, 500+ claims). [ACRE guide \[archive\]](#)

2017–2018 **Postdoctoral Research Fellow**, University of California, Berkeley (BITSS)

- Designed and delivered training curriculum (Git/GitHub, pre-registration, dynamic documents, reproducibility, Mertonian norms) for hundreds of researchers across multiple countries, in collaboration with Prof. Edward Miguel.
- Co-authored IDB report on transparent and reproducible research, adopted as guidance for IDB research staff.

- 2011–2012 **Junior Economic Advisor**, Ministry of Finance, Chile
- Produced time-sensitive economic analyses on labor, social, and education policy for the Minister's office; quantified budget impact of national policies informing multi-billion-dollar allocations.
- 2010–2011 **Research Associate**, Universidad de Chile, Department of Economics
- Impact evaluation of national financial aid program using Regression Discontinuity on administrative records; identified small effects on enrollment and graduation, informing design of follow-up programs.
- 2006–2008 **Economist**, Ministry of Planning, Chile
- Co-led impact evaluation of *Chile Solidario*, the national anti-poverty program, using Difference-in-Differences and matching on administrative + survey data.
 - Produced official poverty and inequality estimates (pre- and post-transfer) used in national reporting.

Education

- 2017 **PhD in Public Policy Analysis**, *Pardee RAND Graduate School*
- 2010 **M.A. in Economics**, *University of Pennsylvania*
- 2006 **B.A. & M.A. in Economics**, *Universidad de Chile*

Publications

Peer-reviewed

- 1 - “Reproducibility and robustness of economics and political science research”; Brodeur, A., Mikola, D., Cook, N., Fiala, L., ..., *Hoces de la Guardia, F.*, ...; *Nature* 652, 151–156, April 2026. doi.org/10.1038/s41586-026-10251-x [preprint]
- 2 - “Promoting reproducibility and replicability in political science”; Brodeur, A., Esterling, K., Ankel-Peters, J., Bueno, N., Desposato, S., Dreber, A., Genovese, F., Green, D., Hepplewhite, M., *Hoces de la Guardia, F.*, et al.; *Research & Politics*, February 2024. doi.org/10.1177/20531680241233439
- 3 - “Reproduction and replication at scale”; Brodeur, A., Dreber, A., *Hoces de la Guardia, F.*, Miguel, E.; *Nature Human Behaviour - Correspondence*, January 2024. doi.org/10.1038/s41562-023-01807-2
- 4 - “Replication games: how to make reproducibility research more systematic”; Brodeur, A., Dreber, A., *Hoces de la Guardia, F.*, Miguel, E.; *Nature - Comment*, September 2023. doi.org/10.1038/d41586-023-02997-5 [preprint]
- 5 - “A Framework for Open Policy Analysis”, *F. Hoces de la Guardia*, S. Grant, E. Miguel; *Science and Public Policy*, April 2021. doi.org/10.1093/scipol/scaa067 [preprint]
- 6 - “A consensus-based transparency checklist”; Aczel, B., Szaszi, B., Sarafoglou, A., Kekecs, Z., Kucharský, Š., Benjamin, D., ..., *F. Hoces de la Guardia*, ..., & Ioannidis, J. P.; *Nature Human Behaviour*, December 2019. doi.org/10.1038/s41562-019-0772-6
- 7 - “Loss Function-based Evaluation of Physician Report Cards”; *F. Hoces de la Guardia*, J. Hwang, J.L. Adams, S.M. Paddock. *Health Services and Outcomes Research Methodology*; February 2018. doi.org/10.1007/s10742-018-0179-2

- 8 - "Optimizing Variance-Bias Trade-off in the TWANG Package for Estimation of Propensity Scores"; L. Parast, D. McCaffrey, L. Burgette, F. Hoces de la Guardia, D. Golinelli, JNV Miles, BA Griffin; Health Services and Outcomes Research Methodology; December 2017. doi.org/10.1007/s10742-016-0168-2 [preprint]
- 9 - "Better-than-average and worse-than-average hospitals may not significantly differ from average hospitals: an analysis of Medicare Hospital Compare ratings"; SM Paddock, JL Adams, F. Hoces de la Guardia; BMJ Quality & Safety; February 2015. doi.org/10.1136/bmjqs-2014-003405
- 10 - "Evaluating the Chile Solidario Program: Results Using the Chile Solidario Panel and the Administrative Databases", F. Hoces de la Guardia, A. Hojman, O. Larrañaga; Revista Estudios de Economía, July 2011. doi.org/10.4067/s0718-52862011000100006

Working Papers

- "Assessing Reproducibility in Economics Using Standardized Crowd-sourced Analysis", Brodeur, A. (r) Sung, S. (r) Miguel, E. (r) Vilhuber, L. (r) Hoces de la Guardia, F.; NBER Working Paper 33753, 2025. nber.org/papers/w33753

Reports

- "Transparency Assessment of Ex-Ante Economic Analyses in Development Organizations", F. Hoces de la Guardia, USAID November 2022. [archived].
- "Open Policy Analysis for a Progressive Wealth Tax", F. Hoces de la Guardia, E. Saez, S. Stout, G. Zucman; BITSS March 2019 bitss.org/research/opa/projects/progressive-wealth-tax.
- "Best Practices for Transparent, Reproducible, and Ethical Research", F. Hoces de la Guardia, J. Sturdy; Inter-American Development Bank, Technical Note #1635 March 2018 dx.doi.org/10.18235/0001564.
- "How Transparency and Reproducibility Can Increase Credibility in Policy Analysis: A Case Study of the Minimum Wage Policy Estimate", F. Hoces de la Guardia; PhD Dissertation, Pardee RAND Graduate School, 2017. doi.org/10.7249/RGSD394; application to minimum wage analysis.

Grants and Awards

- 2026 Co-PI Coefficient Giving, LLM-based tool to automate the encoding of pre-registered hypotheses from the AEA RCT Registry (1 year, \$150K)
- 2022 Co-PI FAIR/NHH, Reporting Guidelines and Publication Bias (3 years, \$880K)
Co-I NIA/NIH, Research Transparency and Reproducibility Training (3 years, \$750K)
Co-I Templeton Foundation, Fostering Open Science Communities (1 year, \$100K)
- 2019 Co-I Research Grant from Peder Sather Center for Advanced Study. Research project: A Study on Research Transparency Practices
Co-I Research Grant from Arnold Ventures. Accelerating Computational Reproducibility in Economics (2 years, \$500K).

Before and During PhD Berkeley Initiative for Transparency in the Social Sciences, Catalyst Grant. Project: “Disseminating Research Transparency in Peru, Bolivia and Chile” (2017); PRGS Partial Scholarship for Doctorate Studies (2013–2015); Fulbright fellowship for graduate studies (2008–2010); Inter-American award for research on social security, Inter-American Conference of Social Security, Mexico City (2006); Academic performance scholarship, Master in Economics, University of Chile. (2005)

Teaching Experience and Mentoring

Instructor in Semester-long Courses (textbooks in parentheses)

- 2022 *Introduction to Econometrics (EC140)* (Angrist & Pischke 2015), University of California, Berkeley. [Materials here](#).
- 2019–2020 *Introduction to Environmental Economics & Policy (EEP1)* (Mankiw-Micro, Banerjee & Duflo 2019, Kwak 2017), University of California, Berkeley.
- 2007–2010 *Econometrics I* (Stock & Watson), Universidad de Chile (2007, 2008, 2010).

Mentoring

- 2019–2026 Supervised 8 research assistants (at least 50% time for one semester or more)
- 2019–2026 Supported the job/graduate school search for the following students (placements in parentheses): Viviane Rocha (FAIR/NHH), Emma Ng (California Public Utilities Commission), Keanu Lin (States United Democracy Center), Max Zhang (Berkeley Initiative for Young Americans), Aleksandra Ma (Deloitte), and Polly Pu (Master’s at CMU)
- 2018–2022 Supervised 27 Berkeley students through the Undergraduate Research Assistance Program (25% of student’s time), over 3 years.

Service and Outreach

- Refereeing *Harvard Data Science Review*; *Contemporary Economic Policy*.
- 2017–2026 *Research transparency trainings and guest lectures*, mostly with BITSS (33).
- Op-eds “How the Administration’s Open Government Plan Can Be More Transparent”, *The Hill*, February 2023 (with E. Miguel); “Aprobando la transparencia y rechazando la polarización”, *La Tercera*, August 2022 (with A. Hojman and E. Miguel); “Credibilidad y transparencia en políticas públicas”, *La Tercera*, November 2021 (with E. Miguel).

Additional Information

- Webpage <https://fhoces.github.io/>
- Software: R, Stata, L^AT_EX, Python, SQL. GitHub profile: github.com/fhoces
- Languages: Spanish and English.
- Nationality: U.S. and Chilean

References

- *Edward Miguel*. Professor of Economics, UC Berkeley. emiguel@berkeley.edu

- *Erik Sørensen*. Professor of Economics, NHH Norwegian School of Economics. erik.sorensen@nhh.no
- *Bertil Tungodden*. Professor of Economics, NHH Norwegian School of Economics. bertil.tungodden@nhh.no
- *Chapin White*. Senior Scholar, USC Schaeffer Institute. chapinwh@usc.edu
- *Emi Nakamura*. Professor of Economics, UC Berkeley. enakamura@berkeley.edu
(Teaching Reference)